



PRAVEEN KUMAR S

UAV Systems Engineer | Flight Testing & Integration
Bengaluru, India • +91 6369487995 • praveenkumar.s4038@gmail.com
linkedin.com/in/s-praveen-kumar-

PROFESSIONAL SUMMARY

UAV Systems Engineer with hands-on experience across the full drone development lifecycle — design, assembly, integration, tuning, and flight validation — on platforms ranging from 7-inch FPV drones to a 100 kg payload heavy-lift UAV. Skilled in ArduPilot, INAV, and Mission Planner, with exposure to defence UAV programs developed in collaboration with Indian Army units.

WORK EXPERIENCE

UAV Systems Engineer — Avian Aerospace Pvt. Ltd., Bengaluru	<i>Nov 2025 – Present</i>
- Design, assemble, integrate, and flight-test multiple UAV platforms: high-altitude surveillance drones, logistics UAVs, VTOL systems, counter-drone platforms, and FPV drones. - Configure and tune flight controllers using ArduPilot, INAV, and Mission Planner, covering firmware setup, sensor calibration, failsafe configuration, and PID tuning. - Own end-to-end development of a 7-inch FPV drone — component selection, build, firmware configuration, and flight validation through iterative test flights. - Lead integration and flight validation of a 15-inch UAV, verifying structural integrity and subsystem compatibility across ground and flight tests. - Drive system integration and operational readiness testing of a 30 kg payload logistics UAV through structured ground and flight test campaigns. - Contribute to structural development, system validation, and flight testing of a 100 kg payload heavy-lift UAV currently under active development. - Fabricate carbon-fibre composite sheets via Hand Lay-Up process for lightweight, high-strength UAV airframe structures. - Represented the company at the North Tech Symposium (May 2026), attended by the Defence Minister of India, demonstrating UAV systems built in collaboration with Army units.	
UAV Engineering Intern — Avian Aerospace Pvt. Ltd., Bengaluru	<i>Jul 2023</i>
- Worked on feature detection, 2D orthomosaic map generation, 3D photogrammetry, and autonomous drone exploration workflows. - Performed flight controller calibration and used 3D printing for rapid prototyping of UAV components.	

TECHNICAL SKILLS

Flight Stacks & GCS: ArduPilot, INAV, Mission Planner, flight controller calibration, PID tuning, failsafe configuration
UAV Systems: Multirotor & VTOL platforms, FPV systems, counter-drone systems, payload integration (up to 100 kg), flight testing, ESC/motor/propulsion selection
Manufacturing & Prototyping: Carbon-fibre composite fabrication (Hand Lay-Up), 3D printing, airframe assembly
Mapping & Vision: 2D orthomosaic mapping, 3D photogrammetry, feature detection
Programming & Tools: Python, C, C++, Linux, MySQL

EDUCATION

B.E. Electronics and Communication Engineering (with Honors)	<i>2021 – 2025</i>
Velammal College of Engineering and Technology, Madurai • CGPA: 7.79 / 10	
Higher Secondary Certificate (HSC)	<i>2020 – 2021</i>
Velammal Vidyalaya, Madurai • 83.4%	
Secondary School Leaving Certificate (SSLC)	<i>2018 – 2019</i>
Velammal Vidyalaya, Madurai • 80.4%	

KEY HIGHLIGHTS

- Hands-on experience across five UAV platform classes within first year of full-time work — from 7-inch FPV to 100 kg heavy-lift.
- Direct exposure to defence UAV programs developed with Indian Army units.
- Selected to represent the organization at a national defence technology symposium (North Tech Symposium 2026).